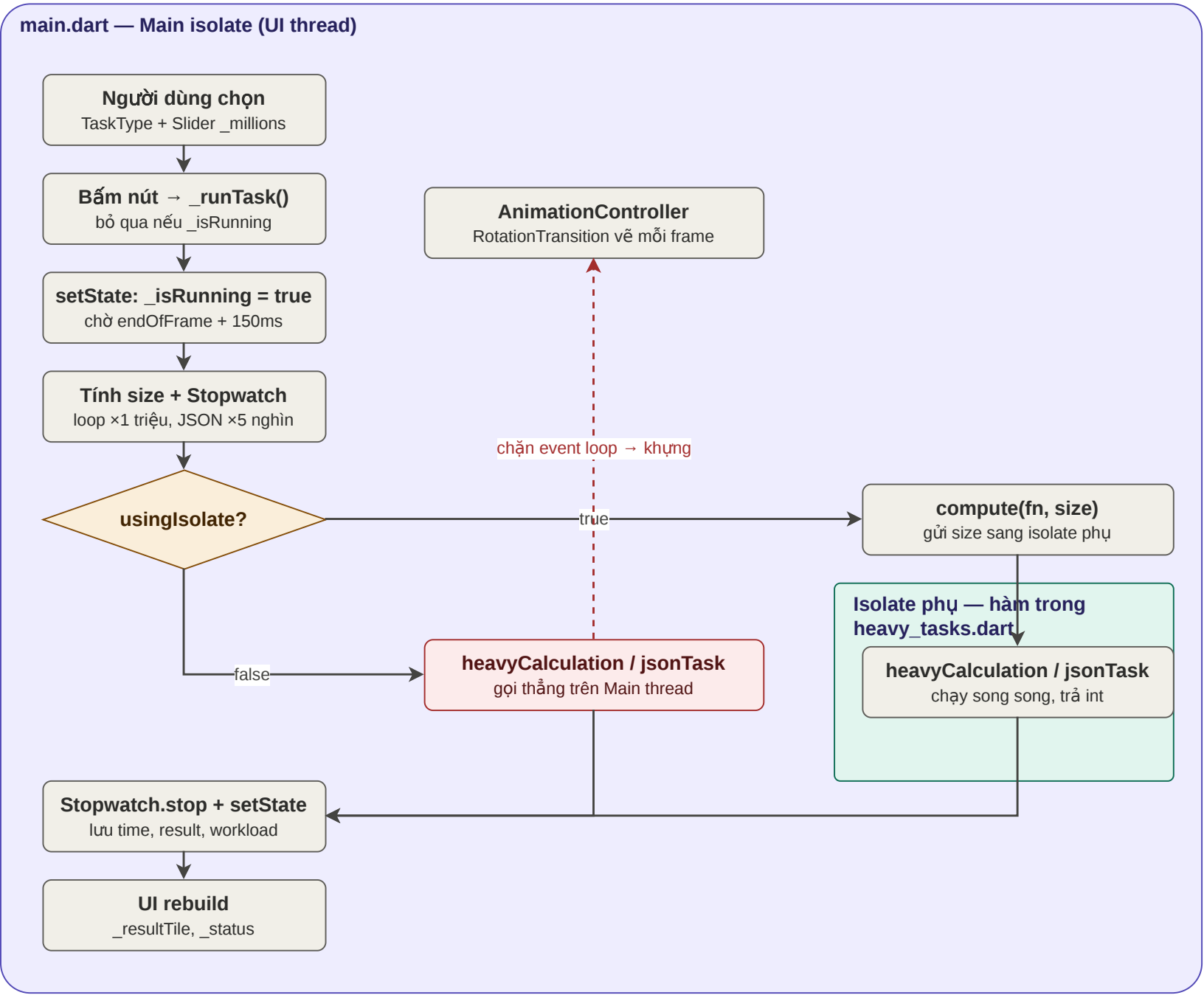
main.dart — Main isolate (UI thread)
Người dùng chọn
TaskType + Slider _millions
Bấm nút → _runTask()
bỏ qua nếu _isRunning
setState: _isRunning = true
chờ endOfFrame + 150ms
Tính size + Stopwatch
loop ×1 triệu, JSON ×5 nghìn
usingIsolate?
true
false
AnimationController
RotationTransition vẽ mỗi frame
chặn event loop → khựng
compute(fn, size)
gửi size sang isolate phụ
Isolate phụ — hàm trong heavy_tasks.dart
heavyCalculation / jsonTask
chạy song song, trả int
heavyCalculation / jsonTask
gọi thẳng trên Main thread
Stopwatch.stop + setState
lưu time, result, workload
UI rebuild
_resultTile, _status